

Phase 3: Unsupervised Patient Stratification & Phenotype Manifolds

A presentation guide evaluating Phase 3 unsupervised patient stratification in the Question 5 Melanoma Pipeline, detailing multi-modal biological clustering, clinical relevance, and key empirical findings across 699 patients.

1. Phase Overview

Analytical Summary & Rationale

- **What is being done:** Grouping 699 melanoma patients into natural biological subgroups using multi-modal machine learning (Gaussian Mixture Models and 1,000-bootstrap consensus clustering) across 9 immunologic and genomic feature axes without outcome bias.
- **Why we are doing it:** Patients respond differently to anti-PD-1 checkpoint immunotherapy. Rather than relying on single gene markers, unsupervised multi-dimensional stratification groups patients by shared tumour microenvironment biology.
- **What question it answers:** What distinct biological phenotypes exist within advanced melanoma, how are patients soft-partitioned across these phenotypes, and how stable are these patient subgroups?

Phase 3 groups patients along 9 multi-dimensional biological traits, combining key inflammatory gene signatures (**TIS** Tumor Inflammation Score, **CYT** Cytolytic Activity), immune cell deconvolution estimates (CD8+ T cells, M1 and M2 Macrophages, Cancer-Associated Fibroblasts **CAFs**), driver mutations (**BRAF** , **NRAS** , **NF1**), and spatial microenvironment metrics.

Rather than forcing patients into rigid binary boxes, Phase 3 assigns continuous soft membership probabilities to reflect intermediate biological states.

2. Upstream Integration & Pipeline Architecture

Phase 3 acts as the central biological foundation of the Question 5 stratification engine, directly integrating discovery modules from Question 1 through Question 4:

- **Integration with Question 1 (Predictive Immune Signatures):** Incorporates the 18-gene Tumor Inflammation Score (**TIS**), cytolytic granzyme/perforin index (**CYT**), and cell-type deconvolution metrics developed in Question 1 to quantify baseline microenvironmental inflammation.
- **Integration with Question 2 (Drug Combination Viability):** Uses targeted driver mutation profiles (**BRAF** V600 , **NRAS** , **NF1** loss) to link transcriptomic immune features with kinase inhibitor sensitivity.
- **Integration with Question 3 (Digital Twin ODE Dynamics):** Provides baseline biological parameters (initial tumour burden, effector T-cell density, macrophage polarization) to initialize 180-day ordinary differential equation (ODE) simulation models for individual patient phenotypes.
- **Integration with Question 4 (Target Nomination):** Maps distinct non-responsive phenotypes to DepMap essentiality targets and LINCX perturbation recommendations (such as CSF1R inhibitors for M2-high tumours and MEK inhibitors for driver-mutated tumours).

3. Why This Matters for Patient Stratification (Q5)

Clinical Relevance & Precision Medicine Rationale

Applying a uniform single-agent checkpoint blockade protocol across all melanoma patients leads to non-response in over 50% of cases. Stratifying patients into distinct biological phenotypes enables:

1. **Avoiding One-Size-Fits-All Therapy:** Identifying which patients possess inflamed microenvironments versus stroma-excluded or driver-mutated barriers.
2. **Quantifying Classification Uncertainty:** Soft probabilistic GMM clustering provides continuous membership scores, alerting clinicians when a patient sits on the boundary between two biological phenotypes.
3. **Guiding Rational Combination Therapies:** Directing non-responsive phenotypes to targeted combination strategies (such as combining anti-PD-1 with stromal modifiers or kinase inhibitors).

4. Key Phase Results & Empirical Findings

Cohort Phenotype Subtype Distribution (N = 699)

Across the full patient matrix (N = 699) and the immunotherapy-treated cohort (N = 326), unsupervised stratification identified four distinct biological subgroups:

Biological Phenotype Subtype	Patient Count (N)	Cohort Share	Key Microenvironmental Profile	Response Rate (ICI Cohort)
Immunosuppressive M2-High	305	43.6%	Depleted T-cell infiltrate, M2 macrophage dominance, and high CAF stromal exclusion walls.	41.4%
Immune Hot	266	38.1%	High TIS and CYT signatures, dense CD8+ T-cell infiltration, inflamed microenvironment.	37.1%
Mutant-Driven	106	15.2%	100% NF1 loss driver mutation subtype with moderate immune background.	37.5%
Immune Cold	22	3.1%	Severe immune desert with uniformly low T-cell infiltration and low signature scores.	66.7%

Visualizing Patient Subgroups in 2D Space

To evaluate how cleanly patient groups separate, Phase 3 generates both linear and non-linear 2D projections:

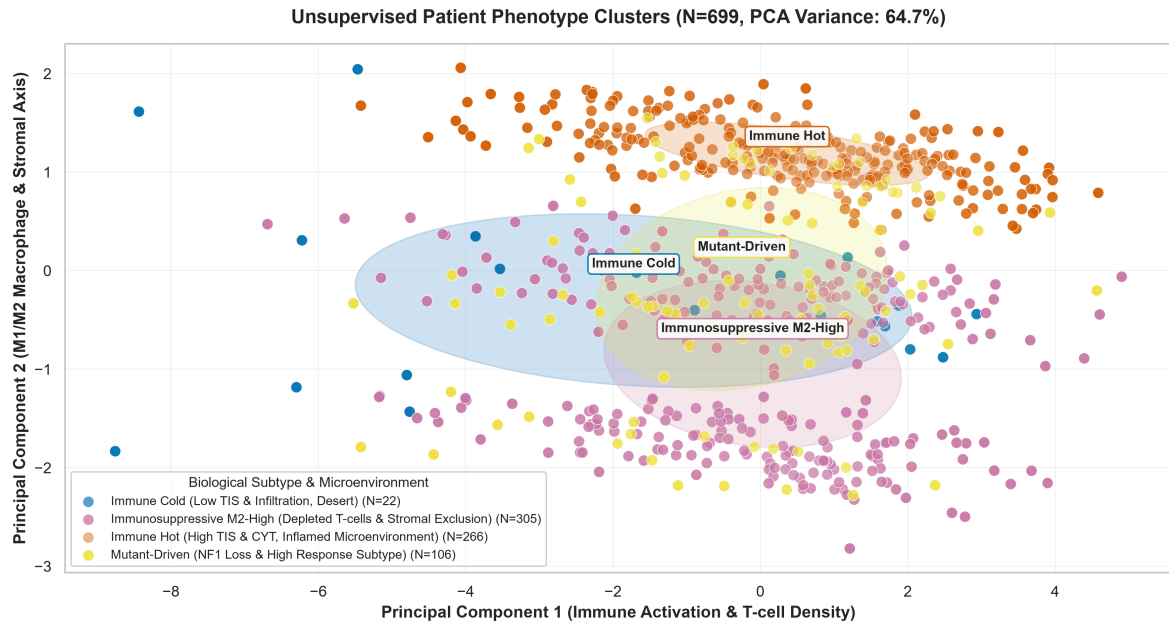


Figure Interpretation: 2D PCA Cluster Projection

- Axis 1 (Horizontal):** Principal Component 1 captures immune activation and lymphocytic T-cell density, separating inflamed hot tumours from cold deserts.
- Axis 2 (Vertical):** Principal Component 2 captures macrophage polarization (M1 vs M2 balance) and stromal CAF exclusion.

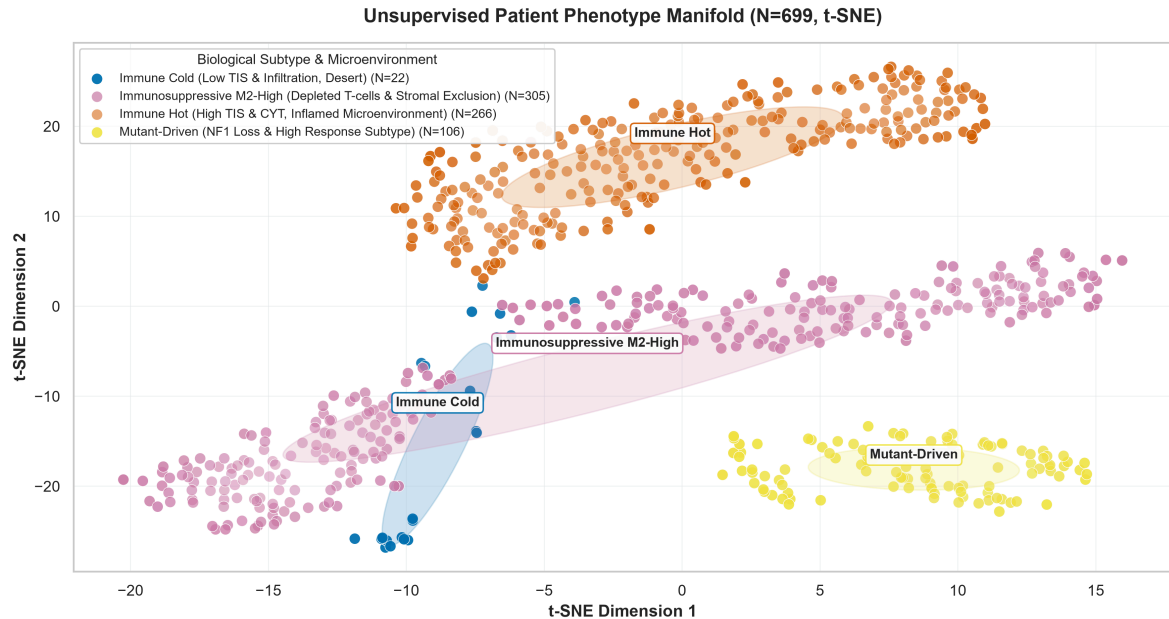


Figure Interpretation: Non-Linear t-SNE Cluster Manifold

- **Non-Linear Topology:** t-SNE preserves local patient neighbourhood relationships across all 9 multi-modal features. Natural within-cluster spread reflects genuine continuous variation rather than rigid artificial boundaries.

Key Biological Takeaways

- **Dominance of Stromal Exclusion:** Data-driven probabilistic clustering reveals that **true immune deserts (Immune Cold, 3.1%) are rare** in advanced melanoma. Instead, **stromal exclusion (M2-High, 43.6%) is the primary non-hot phenotype**, where T cells are trapped by fibrotic fibroblast walls.
- **Continuous Biological Gradients:** Patient microenvironments form a continuous spectrum rather than isolated islands. Assigning soft probability scores allows precise monitoring of hybrid patient states.

5. Limitations & Future Directions

Summary of Methodological & Biological Limitations

1. **Bioinformatic Spatial Proxies:** Spatial distance ratios (`Spatial_CD8_CAF_Distance_Ratio`) are calculated from transcriptomic cell deconvolution estimates rather than direct micrometer measurements on physical tissue slides.
2. **High-Dimensional Parameter Scale:** Fitting full covariance matrices for 4 clusters across 9 features requires estimating 180 parameters. Expanding to 20+ features risks parameter inflation without covariance shrinkage.
3. **Asymmetric Subgroup Sizes:** True immune cold deserts comprise a small minority of cases (N = 22, 3.1%), requiring specialized sampling methods during predictive model training.

Strategic Future Roadmap

- **Physical Spatial Slide Validation:** Validating transcriptomic distance ratios using physical 10x Visium, Xenium, or multiplex immunofluorescence (mIF) tissue slides.
- **1,000-Bootstrap Consensus Stability:** Validating that 4 clusters represents the optimal mathematical elbow (Delta Area score = 0.1499) across 1,000 randomized patient and feature resamplings.
- **Self-Tuning Graph Manifolds:** Implementing adaptive local distance metrics for Spectral Manifold Clustering to automatically capture complex non-linear tumour shapes.